不蒸馏也超强？揭秘端到端文档OCR模型 POINTS-Reader

腾讯开源 腾讯技术工程 2025年9月12日 02:36 广东 9人



目前论文已经被 EMNLP 2025 主会录取，同时模型已经开源

论文: https://huggingface.co/papers/2509.01215

Github: https://github.com/Tencent/POINTS-Reader

HuggingFace: https://huggingface.co/tencent/POINTS-Reader

Demo: https://huggingface.co/spaces/prithivMLmods/POINTS-Reader-OCR

在论文投稿后，我们利用文章中提出的方法持续对模型进行优化，当前在 OmniDocBench 上取得了领先的优势。

Method Type	Methods	Overall$^{Edit}\downarrow$		Text$^{Edit}\downarrow$		Formula$^{Edit}\downarrow$		Formula$^{CDM}\uparrow$	
		EN	ZH	EN	ZH	EN	ZH	EN	ZH
Pipeline Tools	MinerU-pipeline-2.1.1	0.162	0.244	0.072	0.111	0.313	0.581	79.2	48.8
	Marker-1.2.3	0.336	0.556	0.08	0.315	0.53	0.883	17.6	11.7
	Marker-1.7.1	0.296	0.497	0.085	0.293	0.374	0.688	79.0	36.7
	PaddleOCR PP-StructureV3	0.145	0.206	0.058	0.088	0.295	0.535	81.8	52.1
	Mathpix	0.191	0.364	0.105	0.381	0.306	0.454	82.7	64.6
	Docling-2.14.0	0.589	0.909	0.416	0.987	0.999	1	-	-
	Pix2Text-1.1.2.3	0.32	0.528	0.138	0.356	0.276	0.611	78.4	39.6
	Unstructured-0.17.2	0.586	0.716	0.198	0.481	0.999	1	-	-
	OpenParse-0.7.0	0.646	0.814	0.681	0.974	0.996	1	0.106	0
Expert VLMs	**POINTS-Reader-3B**	0.133	0.212	0.062	0.139	0.304	0.465	-	-
	MinerU2.0-2505-0.9B	0.133	0.238	0.045	0.115	0.273	0.506	79.0	50.8
	MonkeyOCR-pro-1.2B	0.146	0.221	0.068	0.118	0.272	0.452	76.7	63.3
	Dolphin	0.356	0.440	0.352	0.440	0.465	0.604	61.6	40.4
	Nanonets-OCR-s	0.283	0.295	0.134	0.231	0.518	0.546	63.2	52.0
	OCRFlux-3B	0.238	0.349	0.112	0.256	0.447	0.716	60.2	31.9
	GOT-OCR	0.287	0.411	0.189	0.315	0.360	0.528	74.3	45.3
	Nougat	0.452	0.973	0.365	0.998	0.488	0.941	15.1	16.8
	Mistral OCR	0.268	0.439	0.072	0.325	0.318	0.495	64.6	45.9
	OLMOCR-sglang	0.326	0.469	0.097	0.293	0.455	0.655	74.3	43.2

介绍

在当今的信息社会，PDF文档是互联网信息传播的重要形式。然而，如何有效地将这些文档内容转换为便于使用的格式一直是一个挑战。这些PDF文档常常蕴含丰富的信息，同时格式也多种多样，包括正文、数学公式和表格等。现有的文档提取方法主要可以分为三类：

1. 传统解析方案：工具如PyMuPDF在解析过程中可能无法准确保留复杂结构，例如数学公式的精确特征，这常导致提取出的文本伴有大量的噪声。
2. pipeline 方案：例如MinerU和Mathpix，这些方法通过集成多个专有模型，可以取得较好的效果。然而，最终的提取效果受限于单个模型的表现，而且优化起来较为繁琐。
3. 端到端方案：端到端方案为文档信息提取提供了一条简洁、高效的路径，它通过直接优化视觉–语言模型来完成任务。然而，这一方案的成败严重依赖于训练数据的质量和规模，而这恰恰是其最大的瓶颈。目前，数据获取方式存在一个两难困境：要么选择成本高昂且难以扩展的人工标注，要么依赖于从现有模型采集数据，但这又面临着继承其原有偏见（bias）的风险。

在端到端方案中，POINTS-Reader提出了一套高度可扩展的数据生成方案，包含两个核心阶段：统一格式预热阶段(Uniform Format Warm-up Stage)和迭代自我改进阶段(Iterative Self-improvement Stage)。这种方法不仅提升了文档提取的效率，还为后续优化提供了坚实的基础。

模型效果

话不多说，我们先来看看模型的提取效果：

单栏 + 数学公式

Local Autoregressive-ness: Contiguous Next-Token Prediction For any integer $k \geq 1$, define

$$\mathbb{I}_{\text{local}}(t,k) = \begin{cases} 1, & \text{if } \{p_{t-i}\}_{i=1}^{k} = \{p_t - i : i = 1, \ldots, k\}, \\ 0, & \text{otherwise.} \end{cases}$$

The *Local AR-ness@k* is then computed as $\text{NTP@}k = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}_{\text{local}}(t,k)$. Local AR-ness measures the tendency to decode the immediate successor of a previously generated token, capturing sequential continuity. It is non-increasing with k, as it becomes harder to maintain longer consecutive spans.

Global Autoregressive-ness: Earliest-First Mask Selection At step t, sort the masked positions $m_{t-1}^{(1)} < m_{t-1}^{(2)} < \cdots < m_{t-1}^{(|M_{t-1}|)}$. Then

$$\mathbb{I}_{\text{global}}(t,k) = \begin{cases} 1, & \text{if } p_t \in \{m_{t-1}^{(1)}, \ldots, m_{t-1}^{(k)}\}, \\ 0, & \text{otherwise.} \end{cases}$$

The *Global FMS-ratio@k* is $\text{FMS@}k = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}_{\text{global}}(t,k)$. Global AR-ness measures the tendency to always unmask the earliest remaining token, capturing a front-to-back filling strategy. Together, these ratios reveal how the model behaves during generation. The ratio is non-decreasing with k, as the criterion becomes easier to satisfy when more early positions are allowed.

A.3 COUPLED GRPO

In this section, we provide a detailed formulation of our Coupled GRPO algorithm. As discussed in §5, we improve upon the baseline GRPO by introducing a coupled sampling scheme for more accurate probability estimation. The complete Coupled GRPO algorithm is presented in Algorithm 1.

Probability Estimation with Coupled Sampling For a given completion sequence o of length L, we select λ timestep pairs $(t, \tilde{t})$ where $t + \tilde{t} = T$. For each pair, we create two complementary masks M_t and $M_{\tilde{t}}$ defined as binary vectors in $\{0,1\}^L$, such that:

$$M_t \vee M_{\tilde{t}} = \mathbf{1}, \quad M_t \wedge M_{\tilde{t}} = \mathbf{0}, \tag{12}$$

where $\vee$ and $\wedge$ denote element-wise OR and AND, respectively, and $\mathbf{1}, \mathbf{0} \in \{0,1\}^L$ are the all-ones and all-zeros vectors. The probability estimation for token o^k (also marked as x_0) is then computed as:

$$\pi_\theta(o^k | c, o^k_{<T}) = \frac{1}{\lambda+1}\left[\sum_{t+\tilde{t}=T}^{\lambda} [\mathcal{L}_t(x_t) + \mathcal{L}_{\tilde{t}}(x_{\tilde{t}})] + \mathcal{L}_T(x_T)\right]. \tag{13}$$

$\mathcal{L}_t$ is the loss term from Eq (11) at timestep t. In detail, we have $\mathcal{L}_t(x_t) = M_t \cdot \frac{1}{t} \cdot \text{CE}(x_t, x_0)$ where CE stands for the cross entropy loss.

Analysis The coupled sampling scheme provides several benefits: (i) **Full Coverage**: Each token is guaranteed to be evaluated exactly once in each coupled pair, ensuring complete coverage of the sequence. (ii) **Reduced Variance**: By evaluating each token under realistic partial-masking contexts, we reduce the variance in probability estimates compared to full masking. (iii) **Computational Efficiency**: The coupled sampling requires only two additional forward passes per update compared to the d1 (Zhao et al., 2025) baseline when $\lambda = 1$. The variance reduction can be formally quantified in the next section, §A.4.

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单栏 + 表格

Table 4: Raw results on coding tasks for LLMs and dLLMs in 7/8B scale. * denotes that the results are collocated from public reports instead of evaluating by ourselves. BigCodeBench has completion (C) and instruction (I) version of template during evaluation. +SFT means we conduct the same instruction tuning with DiffuCoder-Instruct.

Model	HumanEval		MBPP		BigCodeBench (C)		BigCodeBench (I)	
	-	Plus	-	Plus	Full	Hard	Full	Hard
Base Models								
Qwen2.5-Coder	61.6	51.8	75.9	61.4	46.1	16.2	40.2	14.2
OpenCoder*	66.5	63.4	79.9	70.4	40.5	9.5	–	–
LLaDA	35.4	30.5	50.1	42.1	18.9	4.1	–	–
Dream	56.7	50.0	68.7	57.4	23.6	4.1	–	–
DiffuCoder (Stage 2 65B)	67.1	60.4	74.2	60.9	40.2	12.8	1.8	0.7
- Stage 1 65B	39.0	31.7	48.4	38.3	–	–	–	–
- Stage 1 720B	31.1	23.3	38.8	31.3	–	–	–	–
- Stage 2 16B	66.5	61.0	71.9	57.1	36.7	8.8	–	–
Instruct Models								
Qwen2.5-Coder-Instruct	90.2	85.4	83.9	72.0	50.7	21.6	42.2	18.2
Qwen2.5-Coder+SFT	82.9	75.6	80.1	66.1	46.9	16.2	39.5	14.9
OpenCoder-Instruct*	83.5	78.7	79.1	69.0	40.3	16.9	–	–
LLaDA-Instruct	35.4	31.7	31.5	28.6	16.5	2.7	14.1	1.4
Dream-Instruct	57.9	53.7	68.3	56.1	10.6	0.7	11.4	2.7
Dream+SFT	56.7	50.6	71.7	58.7	27.4	6.1	25.9	5.4
DiffuCoder-Instruct (5 ep)	72.0	65.2	75.1	61.9	35.7	12.2	34.0	8.8
- 1 epoch	67.1	60.4	75.7	61.9	31.1	8.1	32.0	6.1
- 1 ep flexible padding	65.2	58.5	71.2	59.5	35.4	11.5	32.4	10.1
+ coupledGRPO (1ep)	73.2	68.3	78.6	67.5	40.4	10.8	37.5	10.8
+ coupledGRPO (2ep)	70.1	65.2	79.1	65.3	42.8	14.9	39.4	13.5
Commercial Models								
GPT 4o*	90.2	–	82.2	–	49.9	–	–	–
Mercury*	90.0	–	77.1	–	45.5	–	–	–
Gemini Diffusion*	89.6	–	76.0	–	45.4	–	–	–

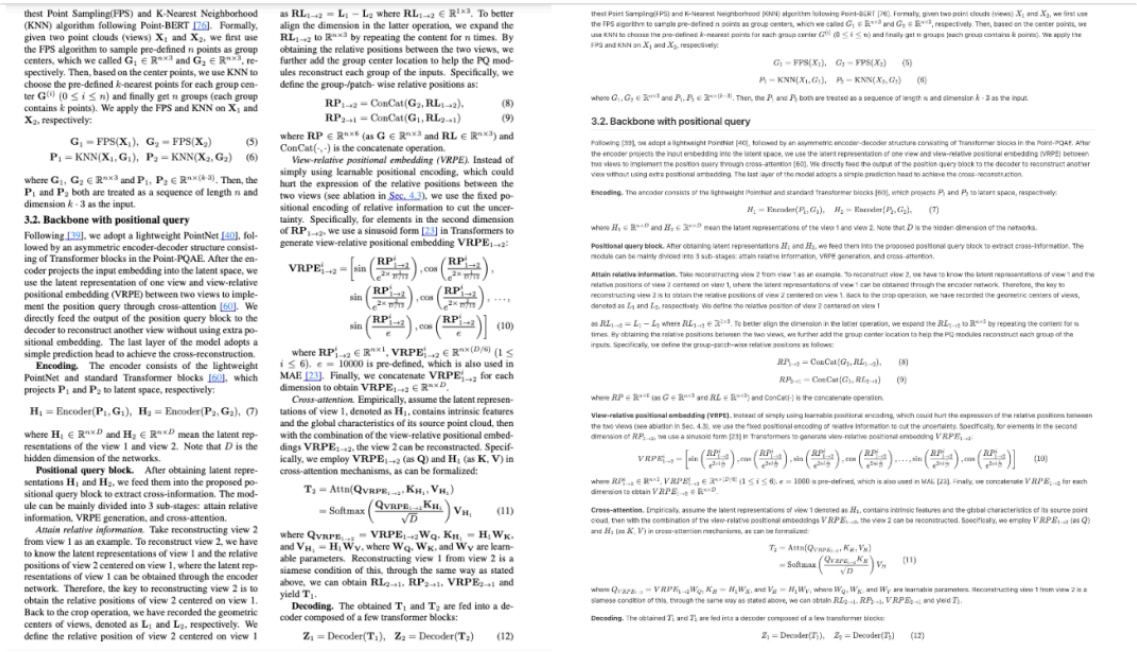
to control the generation length. We compared different SFT strategies for DiffuCoder instruction tuning, including: (i) using a fixed sequence length of 2048 for each sample with a conditional mask; (ii) using a fixed 2048 length but mixing conditional and unconditional training; and (iii) *flexible padding*, where we padded to the maximum sequence length in the current batch instead of a fixed 2048. Finally, we progressively added a conditional mask for the first epoch, considering that Stages 1 and 2 were trained unconditionally, and then trained the remaining 4 epochs using a fixed sequence length of 2048. The SFT code is based on LLaMA-Factory[5]. The training was performed on 8 nodes of 8 H100 GPUs each, using `BF16` and ZeRO2 (Rajbhandari et al., 2020), taking around 24 hours. We used the Adam optimizer with a maximum learning rate of $1e{-}5$ and a linear warmup ratio of 0.1, followed by a cosine decay schedule to 10% of its peak value at the end of training.

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双栏 + 数学公式



双栏 + 表格

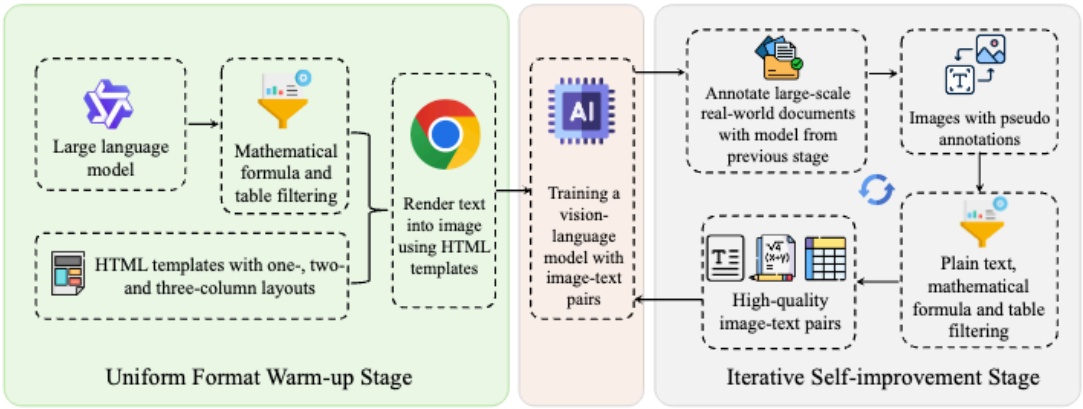


模型特色

1. **简洁性**：POINTS-Reader 是一个非常精简的模型，它保持了与 POINTS1.5 相同的架构，唯一的改动是将 Qwen2.5-7B-Instruct 替换为了 Qwen2.5-3B-Instruct。此外，它的输入和输出都极其简单直接。输入由一个固定的提示（prompt）和一张文档图片组成，输出则是一个单一的字符串（即从文档图片中提取的文本）。模型的输出就是交付给用户的最终结果，无需任何后处理。
2. **性能**：POINTS-Reader 目前支持提取中文和英文两种文档，并取得了非常惊艳的效果，在 OmniDocBench 基准测试上，其英文得分为 0.133，中文得分为 0.212。
3. **高吞吐量**：在当前主流的推理框架（如 SGLang 和 vLLM）中，优化主要集中在大型语言模型（LLM）上。因此，一个大的 ViT（视觉 Transformer）可能会成为一个显著的吞吐量瓶颈，这也是我们选择一个中等规模 ViT（POINTS1.5 中使用的 6 亿参数的 NaViT）的原因。结合我们对 SGLang 的原生支持，我们目前实现了非常高的吞吐量。未来我们也将增加对 vLLM 的支持。

4. **开源技术方案**：在 POINTS-Reader 的论文中，我们提出了一种两阶段的数据增强策略。第一阶段利用自动生成的数据，为模型赋予基础的文档提取能力。在随后的阶段，通过一个持续的自我演进过程来提升模型生成数据的质量。这种第二阶段的自我演进方法具有高度的可扩展性，几乎可以应用于任何模型。

方法



当前，多数端到端模型的训练依赖于私有模型（如GPT-4o）的蒸馏数据。该方法主要面临两大挑战：一是继承教师模型的内在偏见（bias），二是自身迭代受限于外部模型的更新。为解决这些问题，我们提出一种完全不依赖蒸馏数据的两阶段训练方案。

第一阶段：统一格式预热阶段 (Uniform Format Warm-up Stage, UWS)

目标是为模型打下坚实的基础，使其能够处理各种文档元素。

1. 统一输出格式：为了简化模型学习，论文为不同类型的文档元素定义了统一的输出格式：
   - 纯文本：使用Markdown格式。
   - 表格：使用HTML格式（能处理合并单元格等复杂结构）。
   - 数学公式：使用LaTeX格式（行内公式用 `$...$`，行间公式用 `$$...$$`）。
2. 生成大规模合成数据：
   - 使用一个大语言模型（LLM，如Qwen2.5-72B），根据特定的提示（Prompt）生成包含纯文本、含公式的文本、含表格的文本以及多栏布局文本的多样化内容。
   - 对生成的公式和表格进行基于规则的过滤（如语法检查）。
   - 将过滤后的内容使用HTML模板渲染成图像，并与原始文本配对，形成“图像-文本”对。
3. 模型微调：使用这些大规模的、格式统一的合成数据对一个通用的视觉-语言模型（如POINTS-1.5）进行微调。这使模型具备了初步的、格式化的文档元素提取能力。

第二阶段：迭代式自我提升阶段 (Iterative Self-improvement Stage, ISS)

目标是将第一阶段在合成数据上训练的模型，自适应地迁移到真实世界文档上，并持续提升模型和数据质量。

1. 模型标注：使用第一阶段训练好的模型去自动标注一个大规模的真实文档数据集（如DocMatix）。

2. 数据过滤：对模型生成的标注结果应用一套精心设计的、基于规则的过滤策略，以自动验证和筛选高质量的数据：
   - 纯文本过滤：计算模型输出与传统OCR（如PaddleOCR）结果的F1分数，仅保留分数高于阈值（如0.9）的样本，以减少遗漏、幻觉和重复。
   - 表格过滤：检查表格结构的有效性，例如确保每行的单元格数量一致，以去除结构错误的表格。
   - 公式过滤：检查LaTeX公式的语法是否正确，以去除语法错误的公式。
3. 模型再训练：使用经过过滤的高质量真实数据，对模型进行重新训练。
4. 迭代循环：将新训练的模型作为起点，重复执行“标注 -> 过滤 -> 再训练”的过程。通过多轮迭代，模型的转换能力不断提升，同时生成的伪标签数据质量也越来越高。

总结来说，该数据生成方案的核心创新在于：

● 无需蒸馏：完全避免了对昂贵或闭源的教师模型的依赖。

● 自动化闭环：通过“合成数据预热 + 真实数据自我提升”的两阶段设计，构建了一个自动化的数据–模型协同进化闭环。

● 规则化过滤：利用简单但有效的规则对模型自生成的数据进行质量控制，使得模型能在没有人工标注的情况下，利用真实世界数据持续进步。最终我们得到了 POINTS-Reader，在多个基准测试上达到了甚至超过了更大模型的性能。

实验

更强的性能

Evaluation Protocol→		OmniDocBench					Fox
Method	Size	Text↓	Formula↓	Tabl↓	Order↓	Overall↓	Edit Dist↓
Pipeline Methods							
MinerU(Wang et al., 2024a)	-	0.061	0.278	0.18	0.079	0.150	-
Marker(marker, 2024)	-	0.080	0.530	0.619	0.114	0.336	-
Mathpix(mathpix, 2024)	-	0.105	0.306	0.243	0.108	0.191	-
General Vision-language Model							
Qwen2.5-VL-3B(Bai et al., 2025)	3B	0.252	0.429	0.612	0.268	0.390	0.063
Qwen2.5-VL-7B(Bai et al., 2025)	7B	0.144	0.436	0.590	0.154	0.331	0.032
Qwen2.5-VL-72B(Bai et al., 2025)	72B	0.092	0.315	0.341	0.106	0.214	0.027
Expert Vision-language Model							
GOT-OCR(Wei et al., 2024)	716M	0.189	0.360	0.532	0.141	0.287	0.035
Nougat(Blecher et al., 2023)	350M	0.365	0.488	0.572	0.382	0.452	-
Mistral OCR	-	0.072	0.318	0.600	0.083	0.268	-
OLMOCR(Poznanski et al., 2025)	7B	0.097	0.455	0.608	0.145	0.326	-
POINTS-Reader	3B	0.176	0.383	0.335	0.144	0.259	0.023

可以发现，POINTS-Reader 在 OmniDocBench 和 Fox 上都取得了不错的效果，甚至超越了一些私有模型和更大尺寸的模型

消融实验

为了让实验具有更高的可信度，我们对训练 POINTS-Reader 中的每一步进行了详细的消融，下面我们选取几个重点的实验进行展示：

Uniform Format Warm-up 阶段，更好的数据多样性，更好的模型性能

Method	Text↓	Table↓	Formula↓	Order↓	Overall↓
Baseline	0.551	0.652	0.730	0.570	0.626
+Text	0.522	0.641	0.721	0.553	0.609
+Formula	0.513	0.640	0.600	0.530	0.571
+Table	0.495	0.590	0.595	0.523	0.551
+Multi-Column	0.485	0.572	0.511	0.471	0.510

Table 1: **Increasing the diversity of data leads to improved performance (lower is better).** baseline: model fine-tuned with visual instruction tuning data from POINTS-1.5(Liu et al., 2024e). Order: reading order performance. Data is incrementally added from each category.

从上表可以发现，当我们使用更丰富的元素和布局，模型的性能呈现出一个持续变好的趋势。

Iterative Self-improvement 阶段，过滤的手段至关重要

Filtering Strategy	Text↓	Table↓	Formula↓	Order↓	Overall↓
N/A	0.470	0.561	0.514	0.430	0.493
+Text	0.380	0.551	0.501	0.418	0.463
+Table	0.378	0.494	0.500	0.414	0.447
+Formula	0.374	0.492	0.457	0.434	0.439

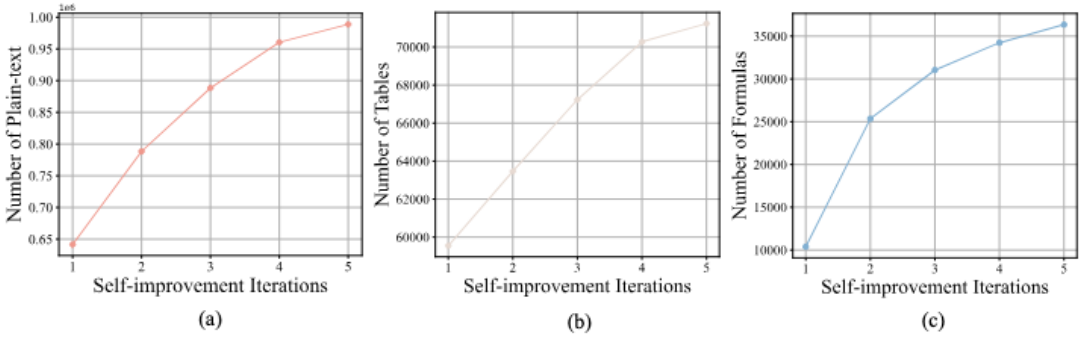
可以发现，在这个阶段对模型生成的数据从各个维度进行过滤，例如表格、公式，可以显著提升数据的质量，从而提升模型的性能。

Iterative Self-improvement 阶段，随着迭代次数多增加，数据的质量和模型的性能也持续提升





从上图可以看出，随着迭代的进行，数据的质量在持续的提升（F1-score 不断增加），但是增长的数据变缓了。同时，随着数据质量的提升，训练得到的模型的性能也不断提升，可以从 OmniDocBench 上不断增长的性能可以看出来



除此之外，我们可以发现，随着迭代次数多增加，我们过滤得到的数据数量也是不断提高，也进一步说明数据质量的提升。

总结

本工作提出了一种全自动化的两阶段数据构建方案。第一阶段，通过统一格式的数据进行预训练，使模型初步掌握文档元素的结构化解析能力。第二阶段，我们引入“标注-过滤-再训练”的迭代循环，驱动模型性能的持续优化。最终，我们成功构建了一个性能卓越的端到端文档解析模型。该方案具有高度的可扩展性，可应用于任何模型高效生产训练数据，并通过自迭代机制实现性能的持续提升。



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阅读 880

留言

写留言

还没有留言，来写首评